

Examiner — Free AI Models & Tools Guide

Goal of This File

This document contains:

- All AI models/tools required for Examiner
- Purpose of each model
- Whether it is FREE
- Recommended usage
- Which phase to use it
- Best architecture decisions

This guide is specifically focused on:

FREE tools and FREE AI models only

for building Examiner.

1. MAIN AI INTERVIEW MODEL

Recommended Model

Google Gemini API

Type:

- Large Language Model (LLM)

Purpose:

- Ask interview questions
- Understand candidate answers
- Generate follow-up questions
- Evaluate answers
- Generate final report
- Conduct complete interview conversation

Free?

✓ YES

Gemini provides a free tier suitable for:

- college projects
 - MVPs
 - development
 - testing
-

Why Gemini Is Best For Examiner

Advantages:

- Powerful reasoning
 - Understands technical subjects
 - Free tier available
 - Fast response
 - Easy API integration
 - Supports long conversations
-

Example Usage

Input:

Subject: Java

Prompt:

Act as a senior Java interviewer.
Ask technical interview questions one by one.
Evaluate answers professionally.

Gemini Output:

What is JVM?

Recommended Phase

Phase 1

Use immediately.

2. SPEECH TO TEXT MODEL

Recommended Model

Whisper

Type:

- Speech Recognition Model

Purpose:

- Convert candidate voice into text

Example:

Candidate Speaks

↓

Whisper Converts Speech → Text

Free?

✓ YES

Open-source model.

You can run locally.

Why Whisper?

Advantages:

- Highly accurate
 - Works with different accents
 - Open-source
 - Easy Python integration
 - Industry standard
-

Recommended Usage

Use for:

- voice interview mode
- real-time candidate speech processing

Recommended Phase

Phase 2

3. TEXT TO SPEECH MODEL

Recommended Tool

Coqui TTS

Type:

- Text-to-Speech Engine

Purpose:

- Make AI interviewer speak

Example:

Gemini Response

↓

Coqui TTS

↓

AI Voice Output

Free?

☒ YES

Open-source.

Why Coqui TTS?

Advantages:

- Natural voice generation
 - Free
 - Local execution possible
 - Good for AI avatar speaking
-

Recommended Phase

Phase 2

4. NLP PROCESSING LIBRARY

Recommended Tool

spaCy

Type:

- NLP Library

Purpose:

- Sentence analysis
 - Keyword extraction
 - Grammar analysis
 - Text processing
 - Communication analysis
-

Free?

 YES

Completely open-source.

Why spaCy?

Advantages:

- Fast NLP processing
 - Easy integration
 - Lightweight
 - Excellent for backend analysis
-

Example Usage

Analyze:

- communication quality
 - sentence structure
 - filler words
 - technical keywords
-

Recommended Phase

Phase 3

5. VOICE CONFIDENCE ANALYSIS

Recommended Tool

Librosa

Type:

- Audio Analysis Library

Purpose:

- Confidence detection
 - Pause detection
 - Speaking speed analysis
 - Voice energy analysis
 - Pitch analysis
-

Free?

✓ YES

Open-source Python library.

Why Librosa?

Advantages:

- Best free audio analysis library
 - Lightweight
 - Perfect for confidence scoring
-

Example Features

Detect:

- nervous pauses
 - unstable voice
 - speaking confidence
 - hesitation frequency
-

Recommended Phase

Phase 3

6. COMPUTER VISION ENGINE

Recommended Tool

OpenCV

Type:

- Computer Vision Library

Purpose:

- Webcam processing

- Face detection
 - Video frame analysis
-

Free?

 YES

Completely open-source.

Why OpenCV?

Advantages:

- Industry standard
 - Fast
 - Huge community
 - Easy webcam integration
-

Example Usage

Analyze:

- candidate face
 - movement
 - focus
 - gestures
-

Recommended Phase

Phase 3

7. FACE & EYE TRACKING MODEL

Recommended Tool

MediaPipe

Type:

- Real-Time Vision Framework

Purpose:

- Face mesh
 - Eye tracking
 - Head pose estimation
 - Body posture tracking
-

Free?

☒ YES

Open-source by Google.

Why MediaPipe?

Advantages:

- Very fast
 - Real-time tracking
 - Easy integration
 - Excellent for interview analysis
-

Example Features

Detect:

- eye contact
 - head movement
 - posture
 - attention level
-

Recommended Phase

Phase 3

8. FACIAL EMOTION DETECTION

Recommended Tool

DeepFace

Type:

- Facial Analysis Framework

Purpose:

- Emotion detection
 - Facial expression analysis
-

Free?

 YES

Open-source.

Why DeepFace?

Advantages:

- Very easy to use
 - Multiple emotion models included
 - Good accuracy
-

Example Features

Detect:

- nervousness
- stress
- happiness

- confusion
 - sadness
-

Recommended Phase

Phase 3

9. FRONTEND ANIMATION SYSTEM

Recommended Tools

React Three Fiber

Purpose:

- 3D interviewer rendering

Free?

☒ YES

Framer Motion

Purpose:

- Smooth UI animation

Free?

☒ YES

Ready Player Me

Purpose:

- AI avatar generation

Free?

☒ YES (basic usage)

Recommended Phase

Phase 1 & 2

10. BACKEND FRAMEWORK

Recommended Tool

Spring Boot

Purpose:

- Main backend server
 - API handling
 - Authentication
 - Session management
 - Interview orchestration
-

Free?

 YES

Open-source.

Why Spring Boot?

Advantages:

- Industry-level backend
 - Scalable architecture
 - Excellent for placements
 - Secure
 - Fast API development
-

Recommended Phase

Phase 1

11. DATABASE

Recommended Database

MongoDB

Purpose:

- Store users
 - Store interview reports
 - Store sessions
 - Store analytics
-

Free?

☒ YES

MongoDB Atlas provides free tier.

Recommended Phase

Phase 1

12. REAL-TIME COMMUNICATION

Recommended Tool

WebSocket

Purpose:

- Real-time interview communication
 - Live chat updates
 - Real-time AI responses
-

Free?

☒ YES

Recommended Phase

Phase 2

FINAL AI ARCHITECTURE

Phase 1 Architecture

```
graph TD; A[React Frontend] --> B[Spring Boot Backend]; B --> C[Gemini API];
```

Features:

- Subject input
- AI questions
- Text answers
- AI evaluation
- Final report

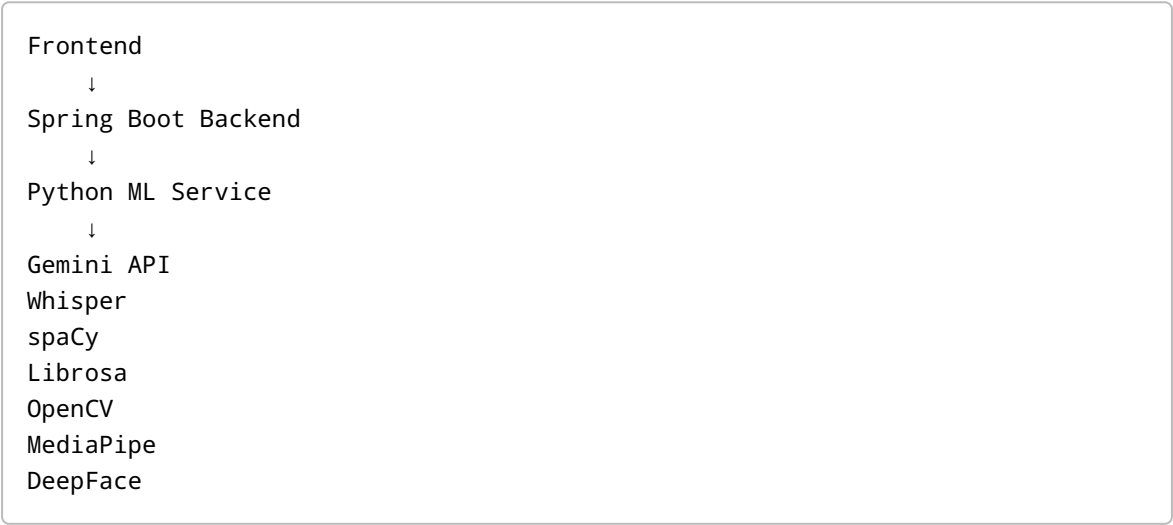
Phase 2 Architecture

```
graph TD; A[Frontend] --> B[Spring Boot Backend]; B --> C[Gemini API]; C --> D[Whisper]; D --> E[Coqui TTS];
```

Features:

- Voice interview
 - AI speaking avatar
 - Real-time conversation
-

Phase 3 Architecture



Features:

- Confidence analysis
- Emotion detection
- Eye tracking
- Communication scoring
- Behavioral analytics

WHAT EACH MODEL DOES

Tool/Model	Purpose
Gemini API	Interview intelligence
Whisper	Speech-to-text
Coqui TTS	AI voice generation
spaCy	NLP analysis
Librosa	Confidence analysis
OpenCV	Webcam processing
MediaPipe	Eye/head tracking
DeepFace	Emotion detection
Spring Boot	Backend server
MongoDB	Database

MOST IMPORTANT UNDERSTANDING

You are NOT building:

A new ChatGPT

You ARE building:

An AI-powered interview experience platform

Your innovation is:

- AI workflow
- interview simulation
- dynamic questioning
- confidence analysis
- interaction system
- realistic interviewer experience

NOT training a giant LLM.

RECOMMENDED DEVELOPMENT ORDER

STEP 1

Build:

- frontend
 - Spring Boot backend
 - Gemini integration
 - text interview system
-

STEP 2

Add:

- voice interaction
 - Whisper
 - Coqui TTS
-

STEP 3

Add:

- OpenCV
- MediaPipe
- DeepFace
- Librosa

FINAL RECOMMENDATION

For Examiner:

Use:

✓ Gemini API ✓ Whisper ✓ Coqui TTS ✓ spaCy ✓ Librosa ✓ OpenCV ✓ MediaPipe ✓ DeepFace

All are:

FREE or provide free usage suitable for student projects

and are perfect for building an advanced AI interviewer system.